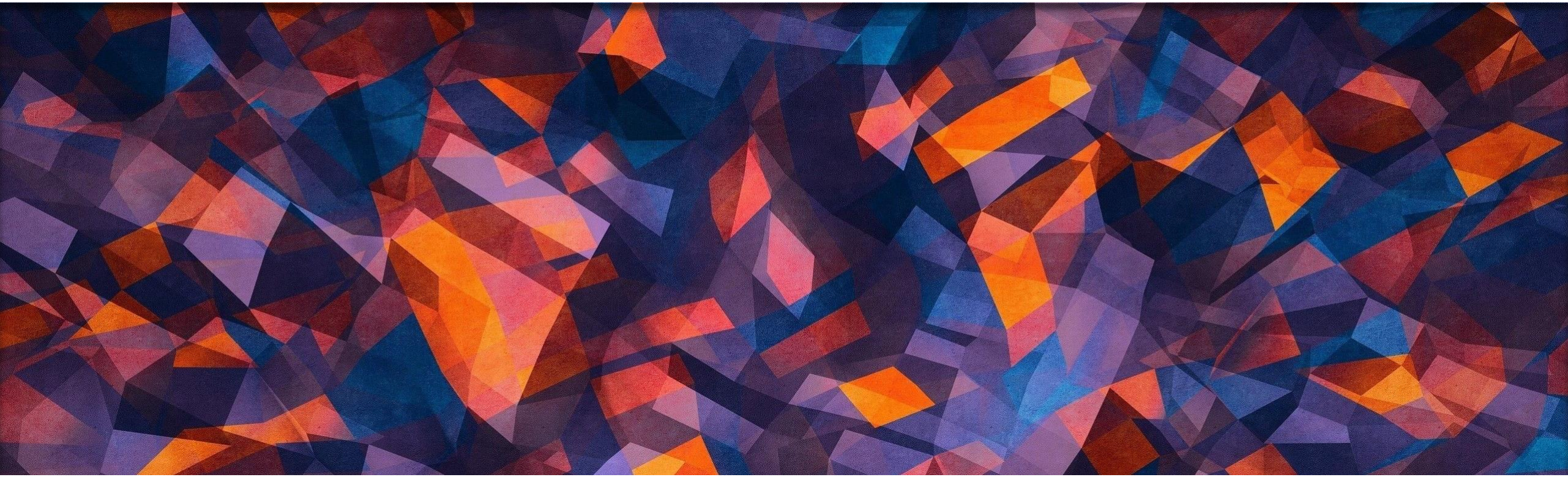
Universität St.Gallen

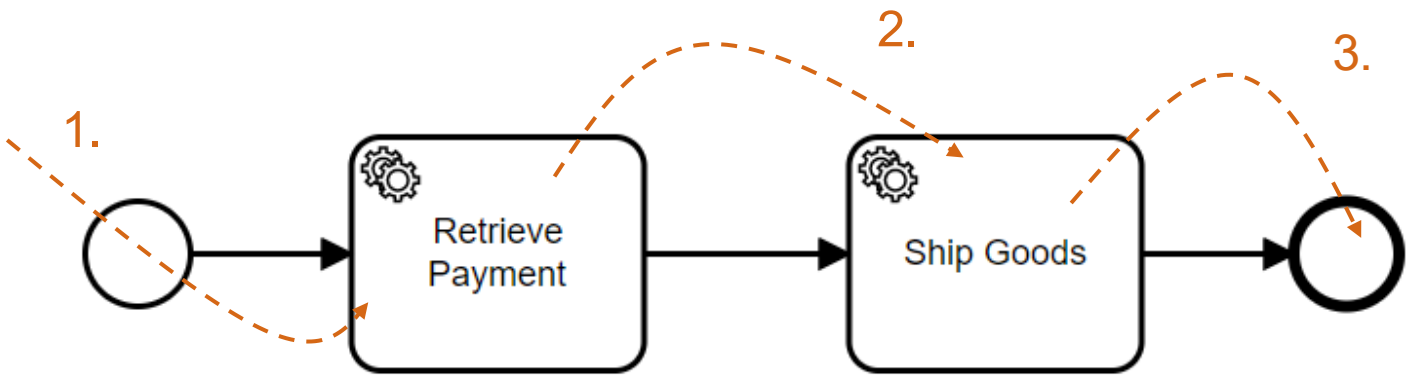


Event-driven and Process-oriented Architectures

Zeebe.io – Example of an Event-driven Architecture

Lecture 6

How Traditional Workflow Engines Work?



The current state of a workflow instance is stored in a database table; engine can leverage guarantees from relational databases

1. INSERT

Workflow instance ID	Current activity	State
2	Retrieve Payment	Running

1. UPDATE

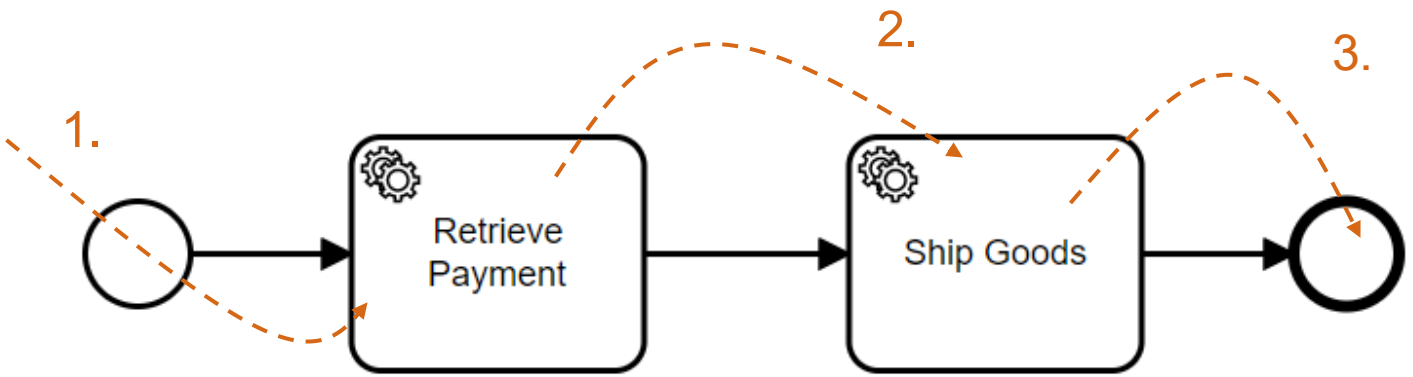
Workflow instance ID	Current activity	State
2	Ship Goods	Running

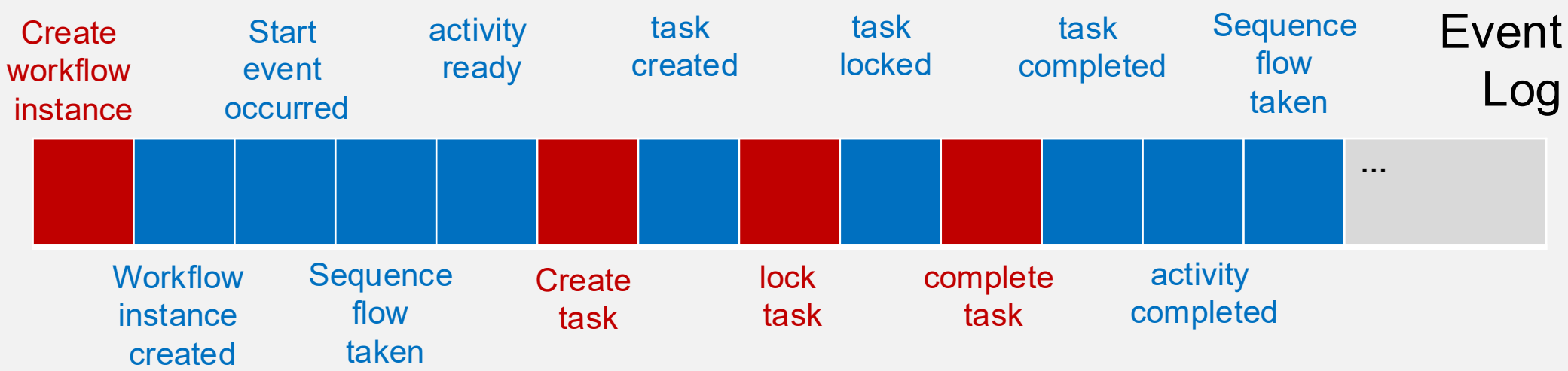
2. UPDATE

Workflow instance ID	Current activity	State
2	Order Delivered	Ended

RDMS

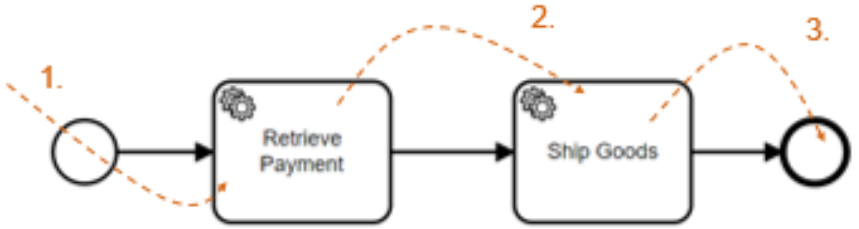
Zeebe.io – An Event-Sourced Engine

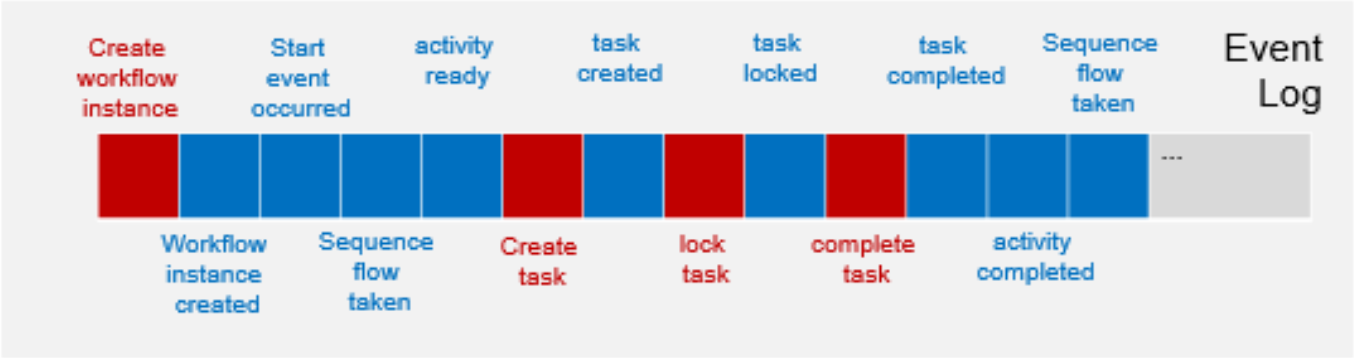




Zeebe.io – An event-sourced Engine

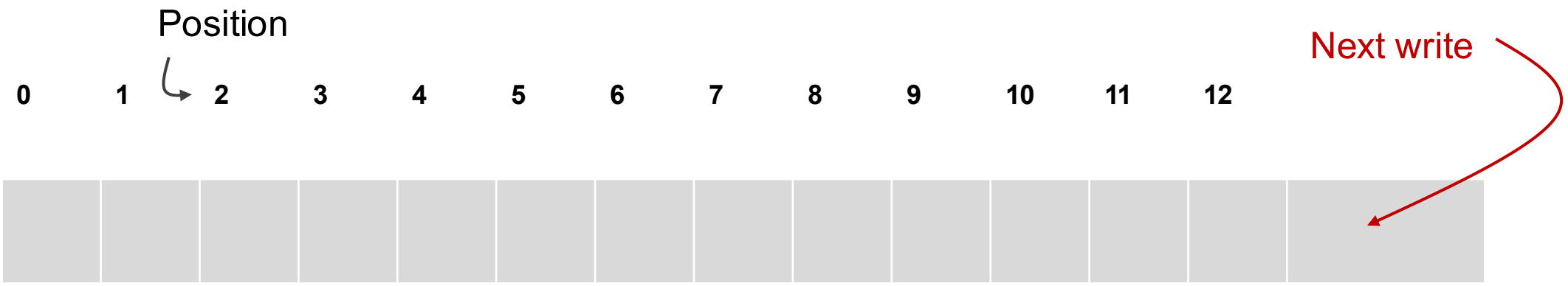
- Zeebe leverages *event sourcing*
- All changes to the workflow state are captured as *events* and stored alongside *commands* in an event log (append only).
- Both events and commands are considered to be *records* in the log.



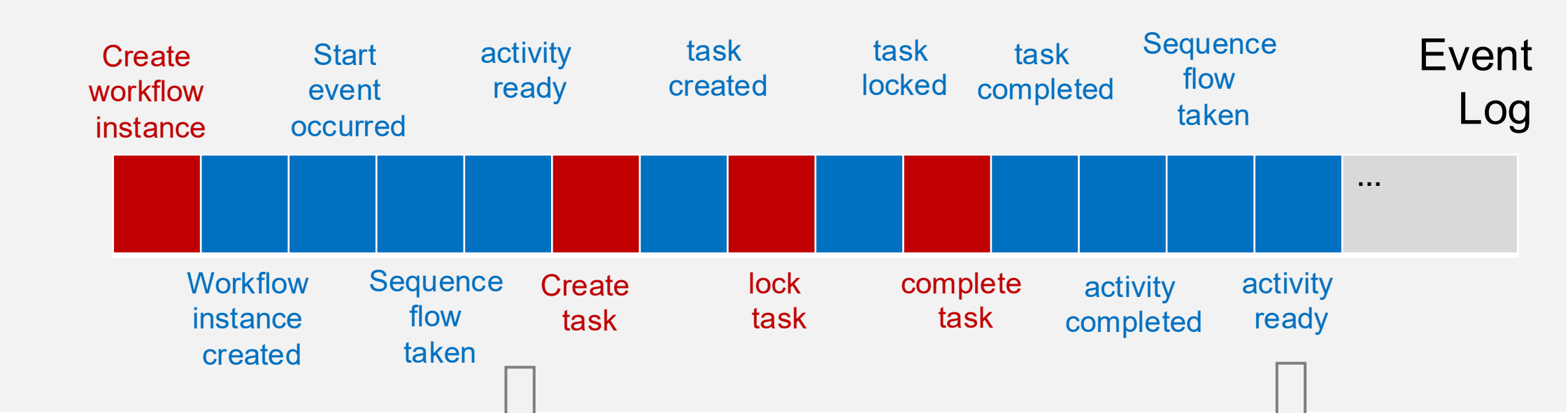


Event Logs are Append-Only

- Event logs are immutable (append-only)
- Append-only logs can be scaled and handled very efficiently



Current State Can be Derived from Events



Old snapshot

Workflow instance ID	Current activity	State
2	Retrieve Payment	Running

New snapshot

Workflow instance ID	Current activity	State
2	Ship Goods	Running

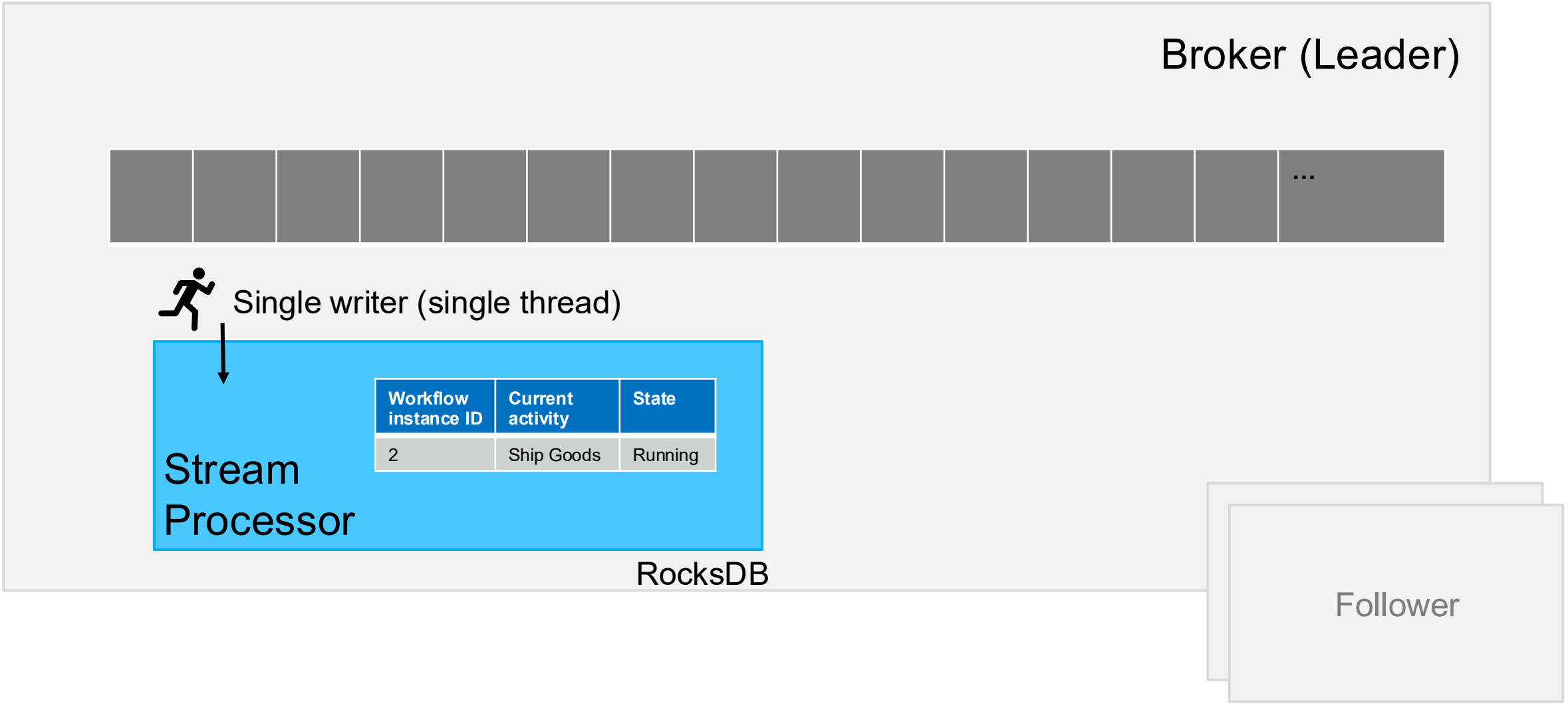
Single Writer Principle

- Zeebe is based on the Single Writer Principle
  - Always exactly one thread that writes to a certain log
- Consequence
  - Clear ordering
  - No locking is needed
  - No deadlock can occur
- Horizontal scalability achieved through partitions (see later)

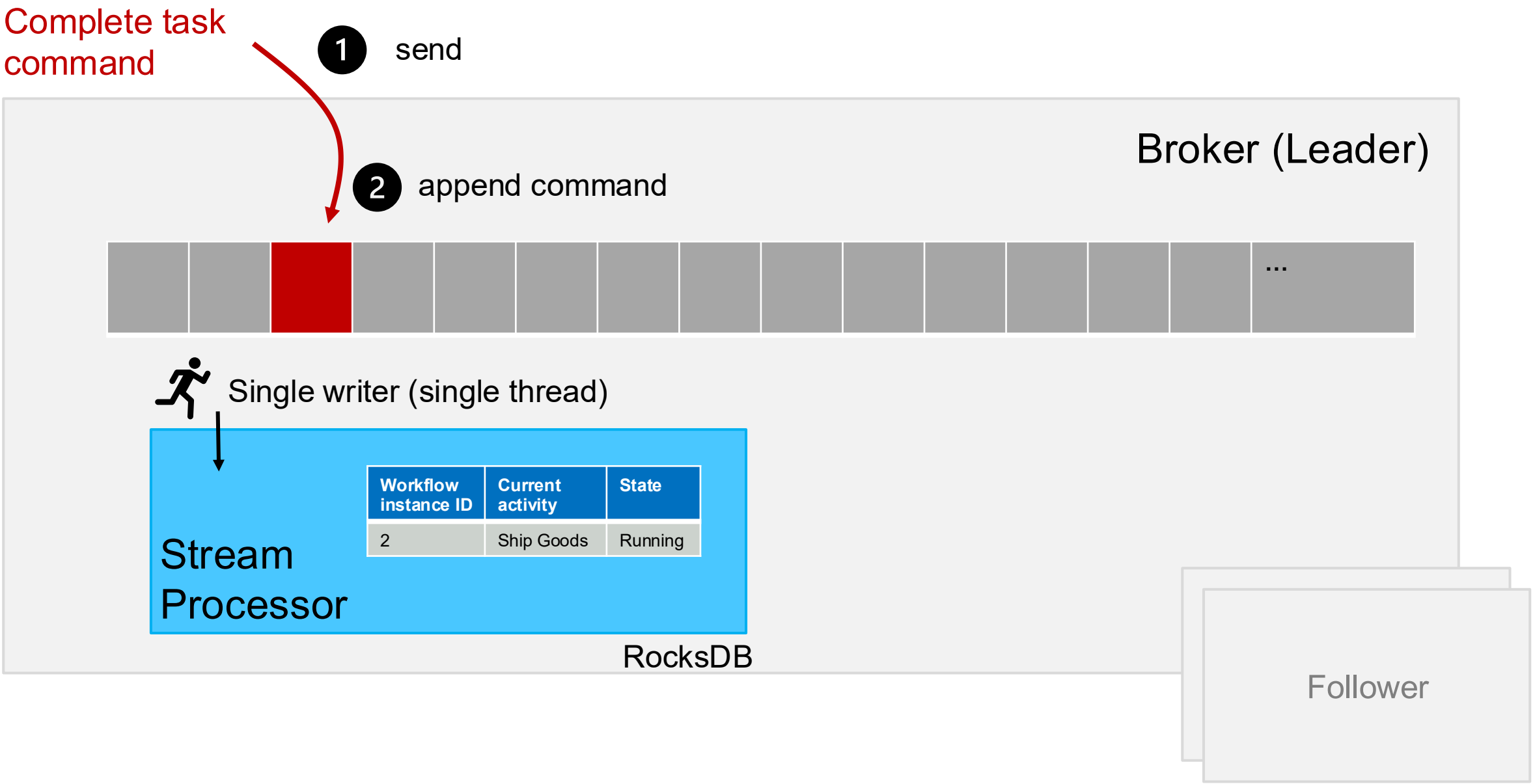
The Event Handling Loop

Complete task command

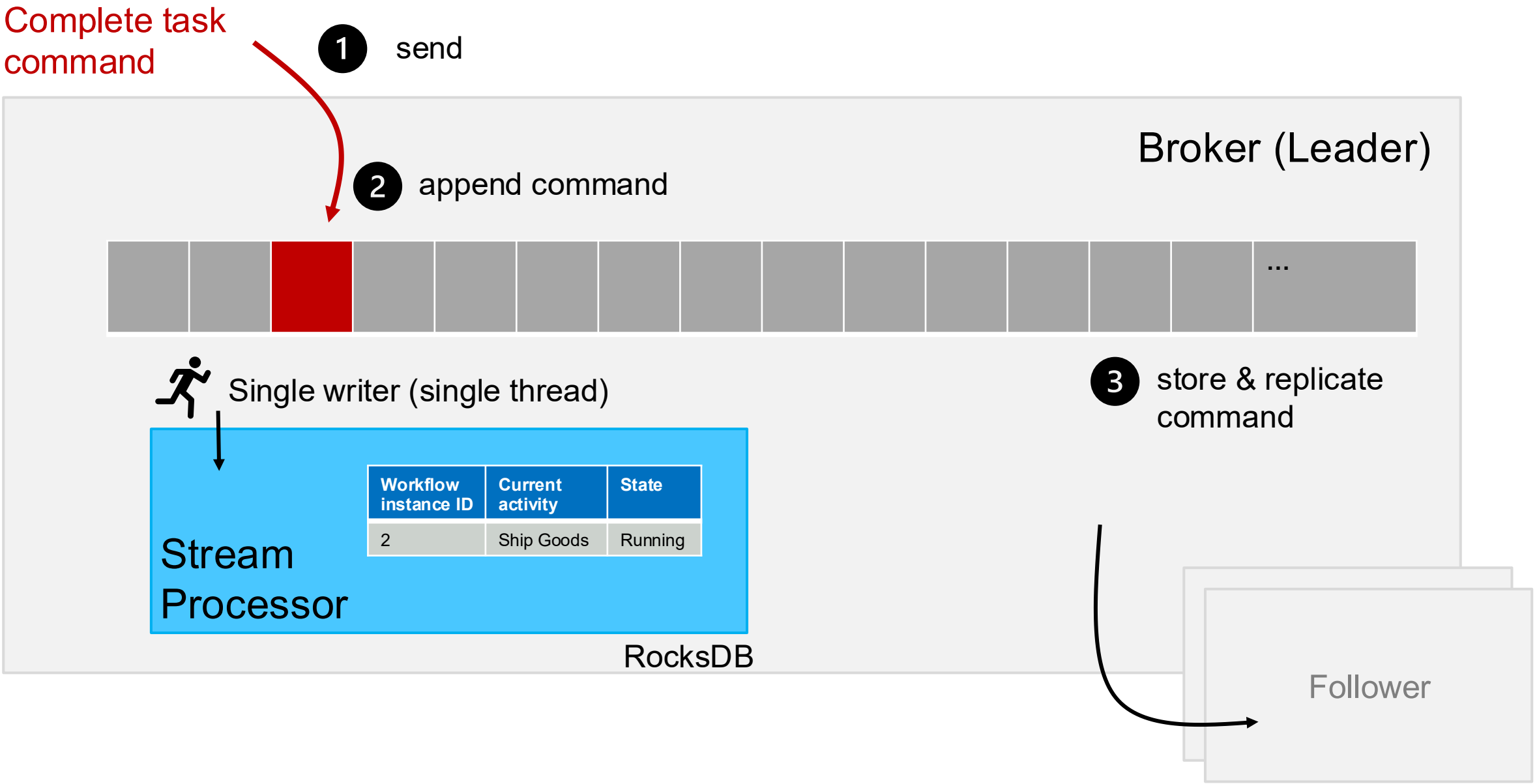
1 send



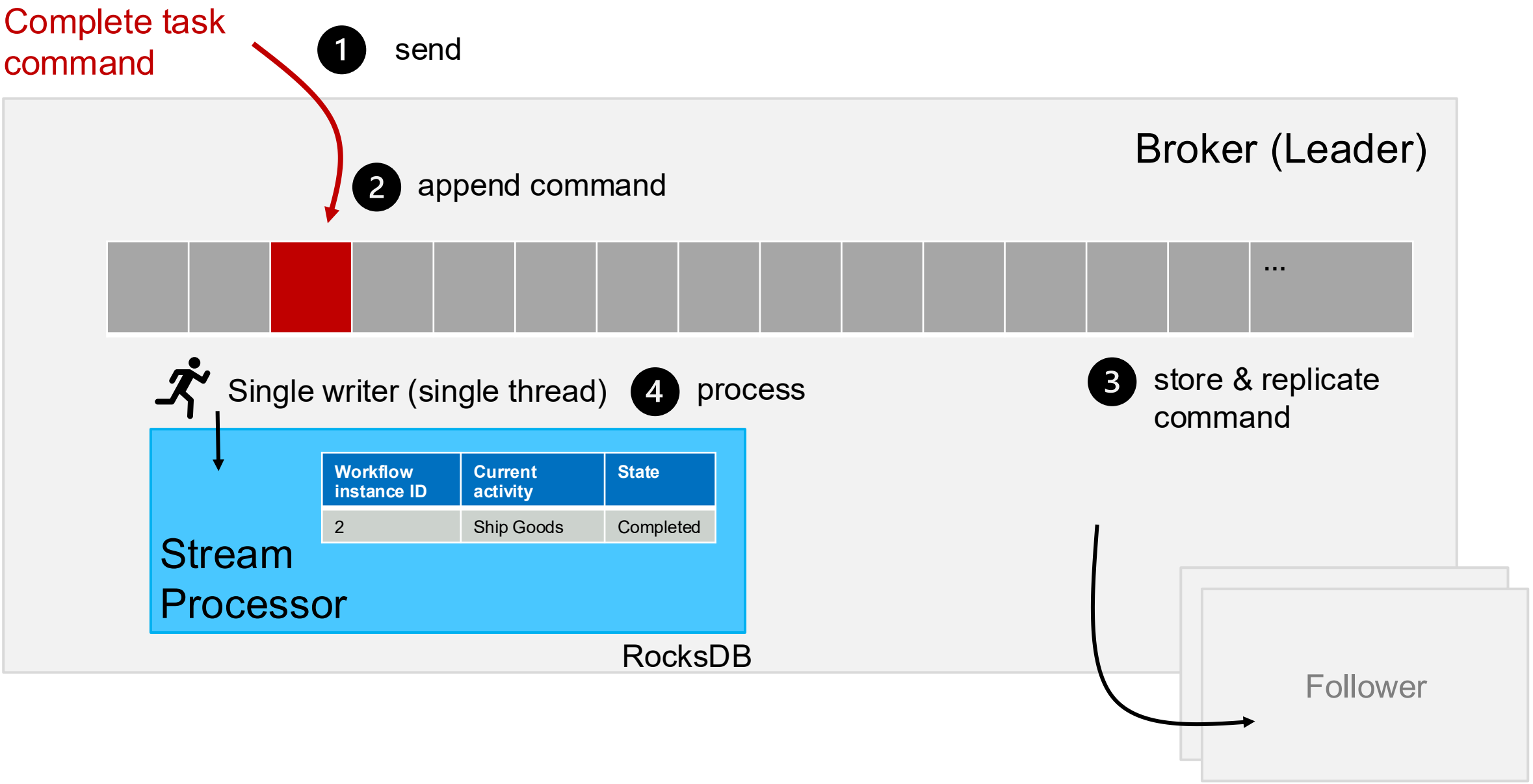
The Event Handling Loop



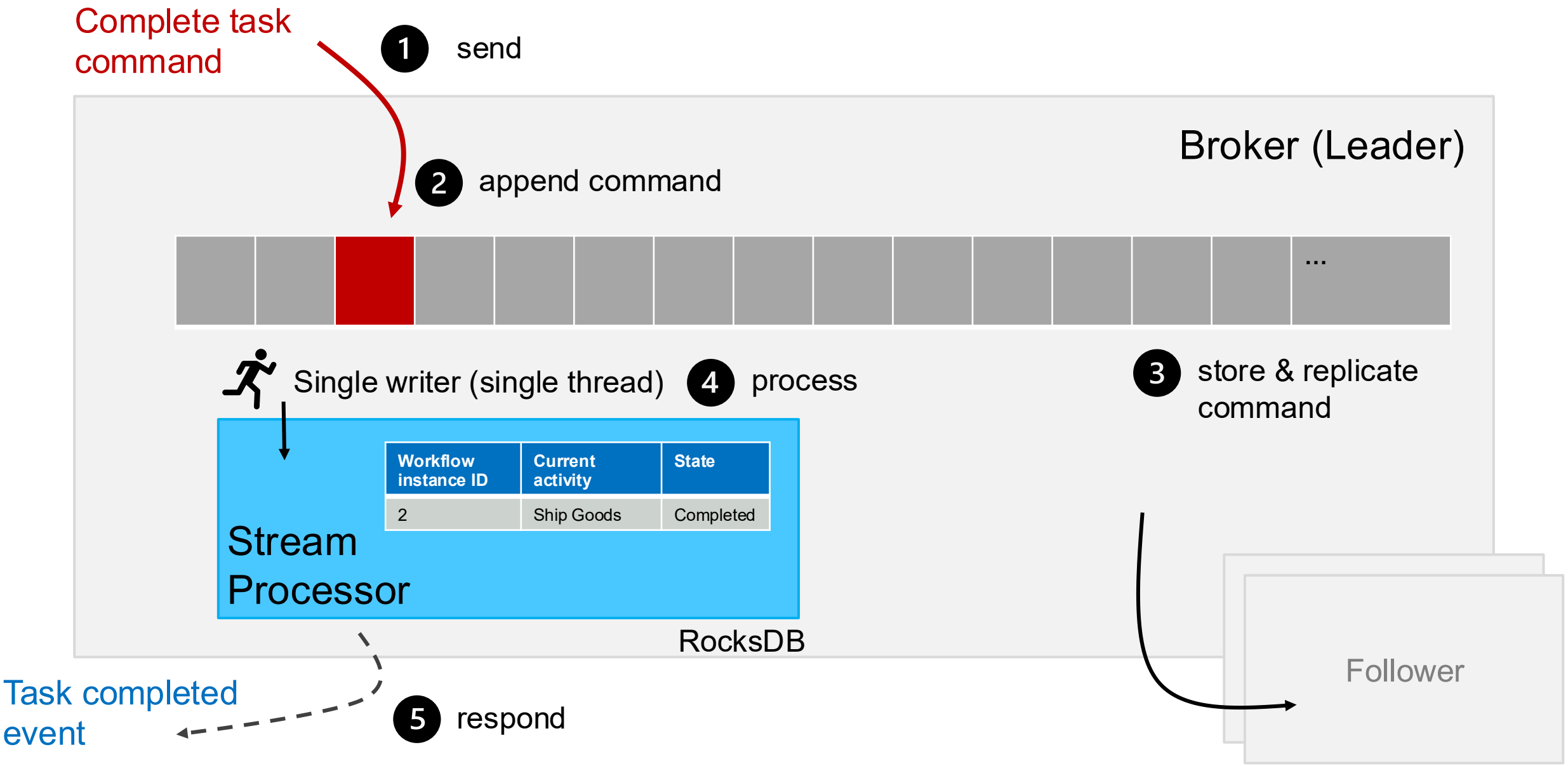
The Event Handling Loop



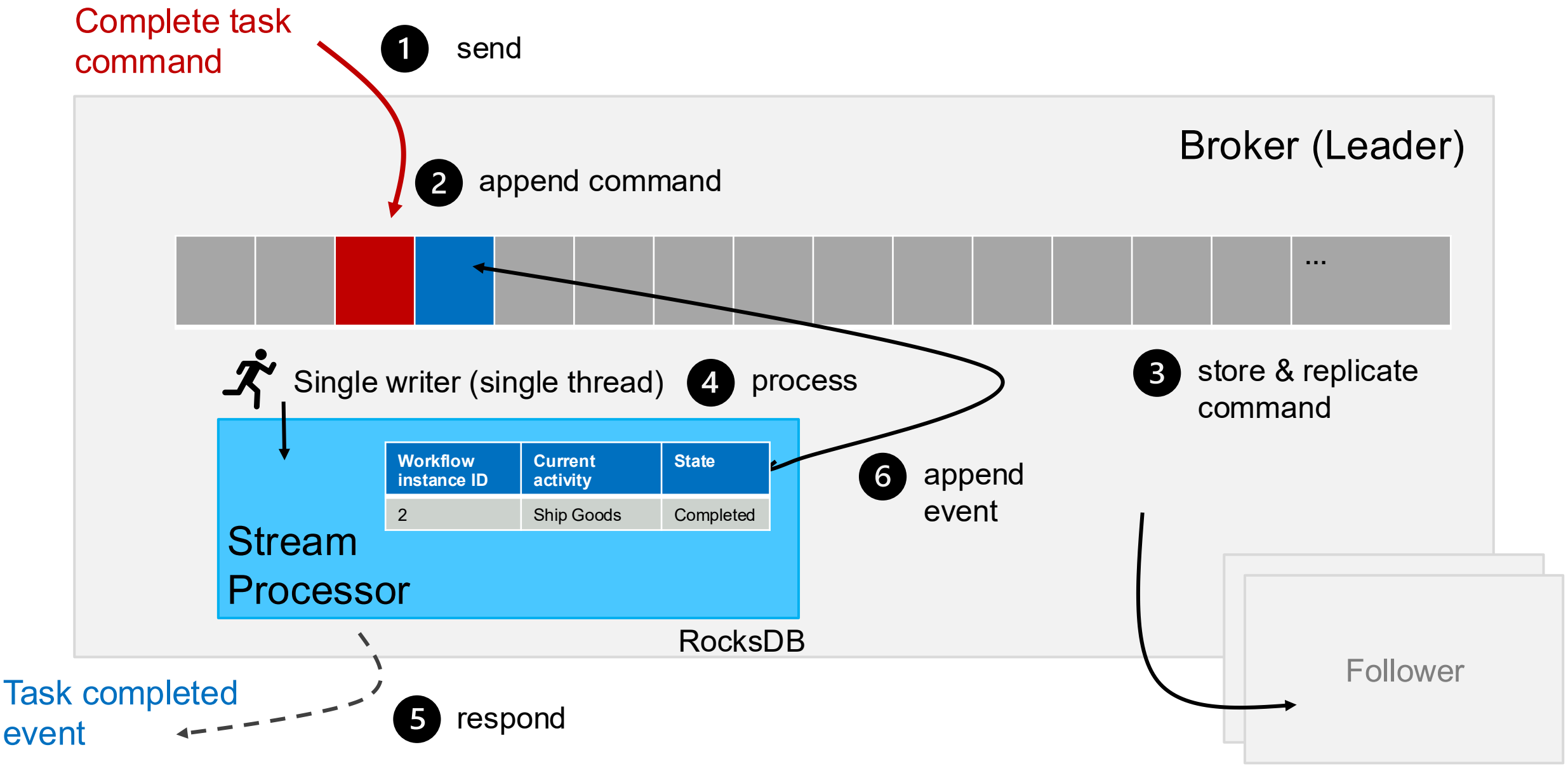
The Event Handling Loop



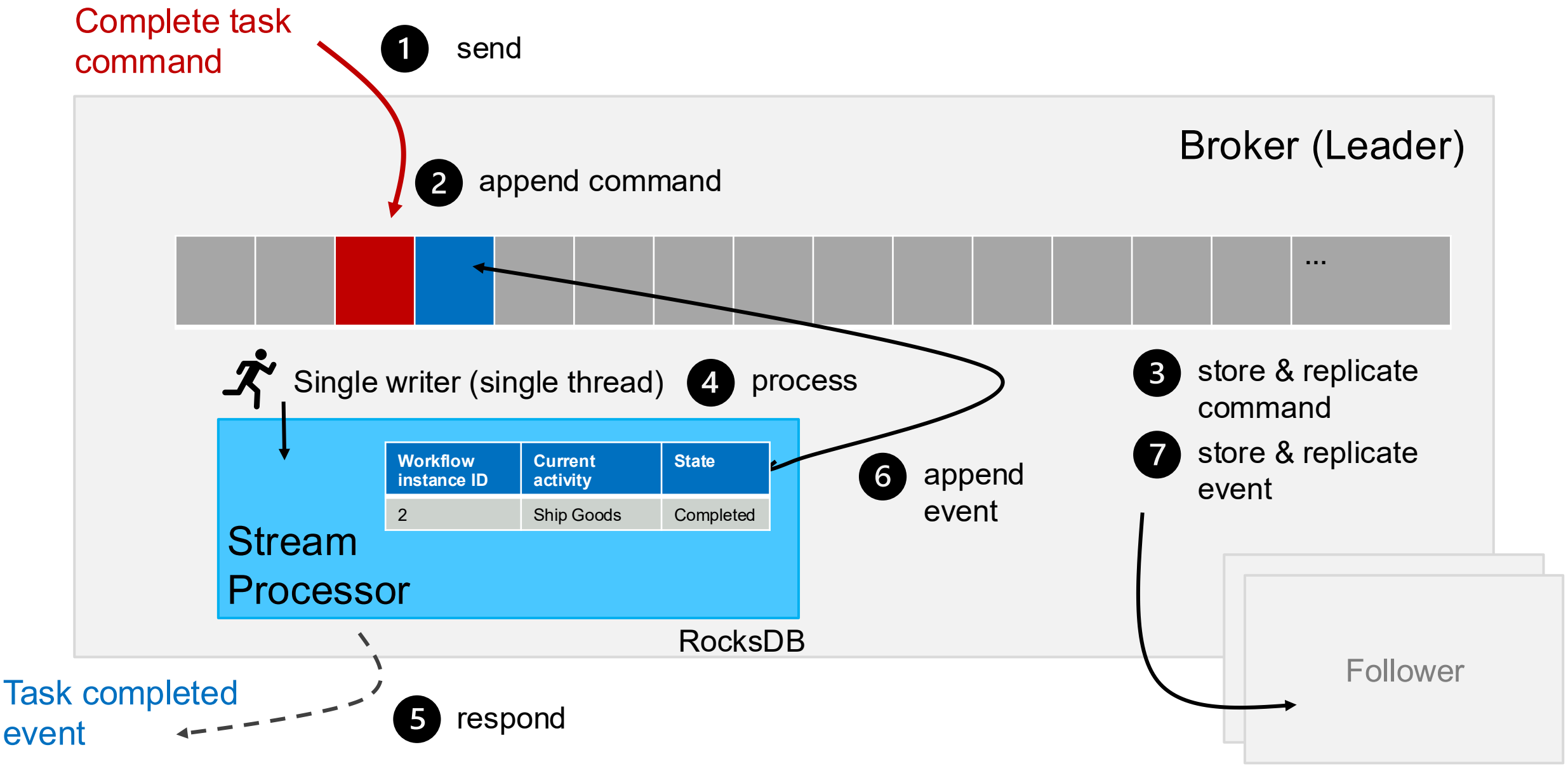
The Event Handling Loop



The Event Handling Loop



The Event Handling Loop



Stream Processing and Exporter

- The append-only log is a **stream of events**
- Zeebe is built around the concept of **processors**, each being a single thread
- One processor implements the **BPMN Engine**
  - understands commands and events semantically and knows what to do next
  - responsible for rejecting invalid commands
- Other class of processors are **exporters**
  - Exporter for Elasticsearch: storage and querying
  - Operate leverages this data to visualize running instances

Log Compaction

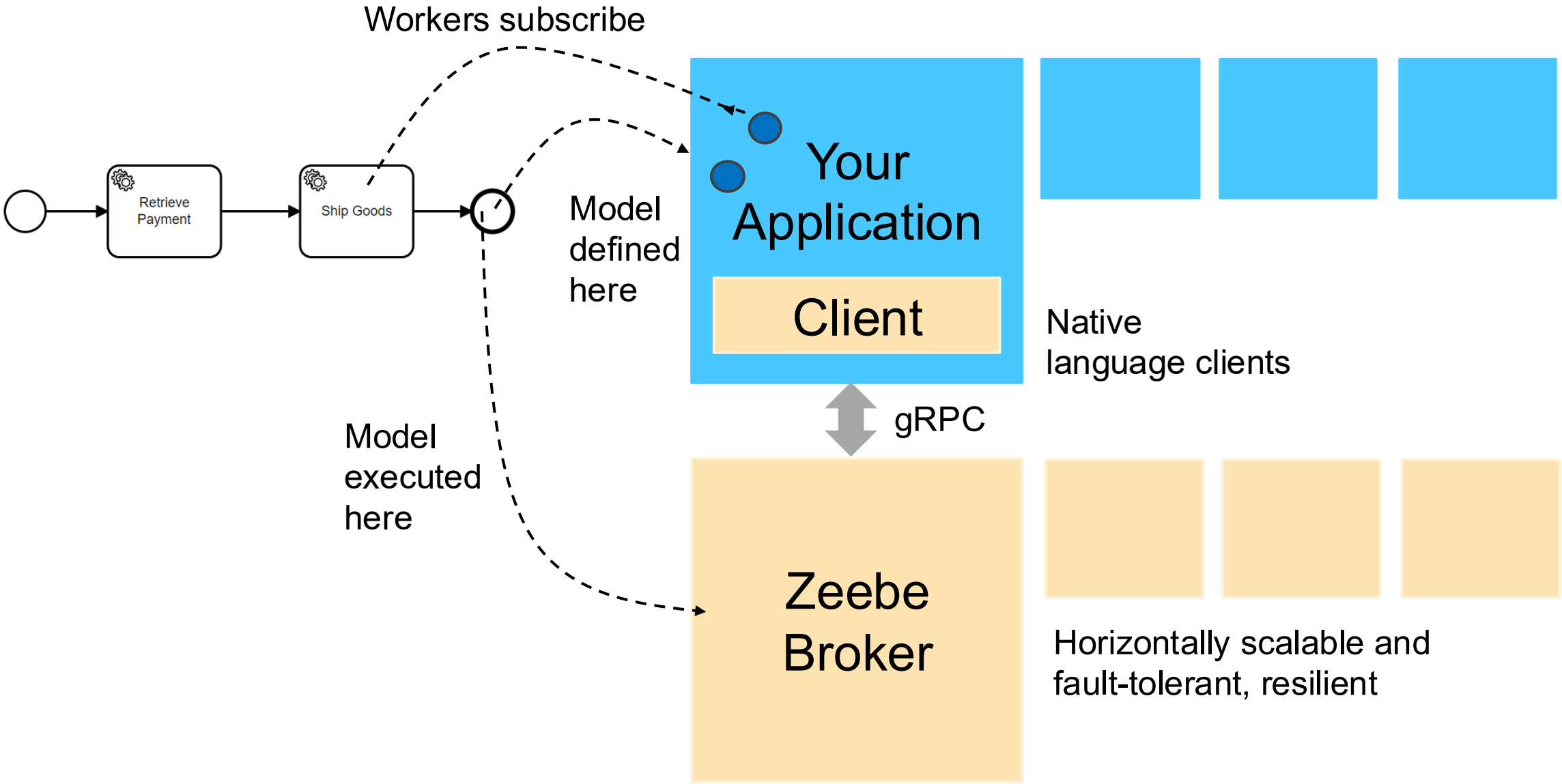
Log Compaction

- Log compaction refers to deleting old data from the log
  - E.g., deleting events for completed workflow instances
  - Complex, since events are scattered all over the place as instances can be long running

Alternative approach

- Events are deleted as soon as they are completely processed (i.e., all stream processors have processed it) and applied to the snapshot
  - allows to keep the log clean without loosing the benefits of an append-only log and stream processing
  - adding stream processors later on and replaying all events from history is not possible (unlike with Kafka)

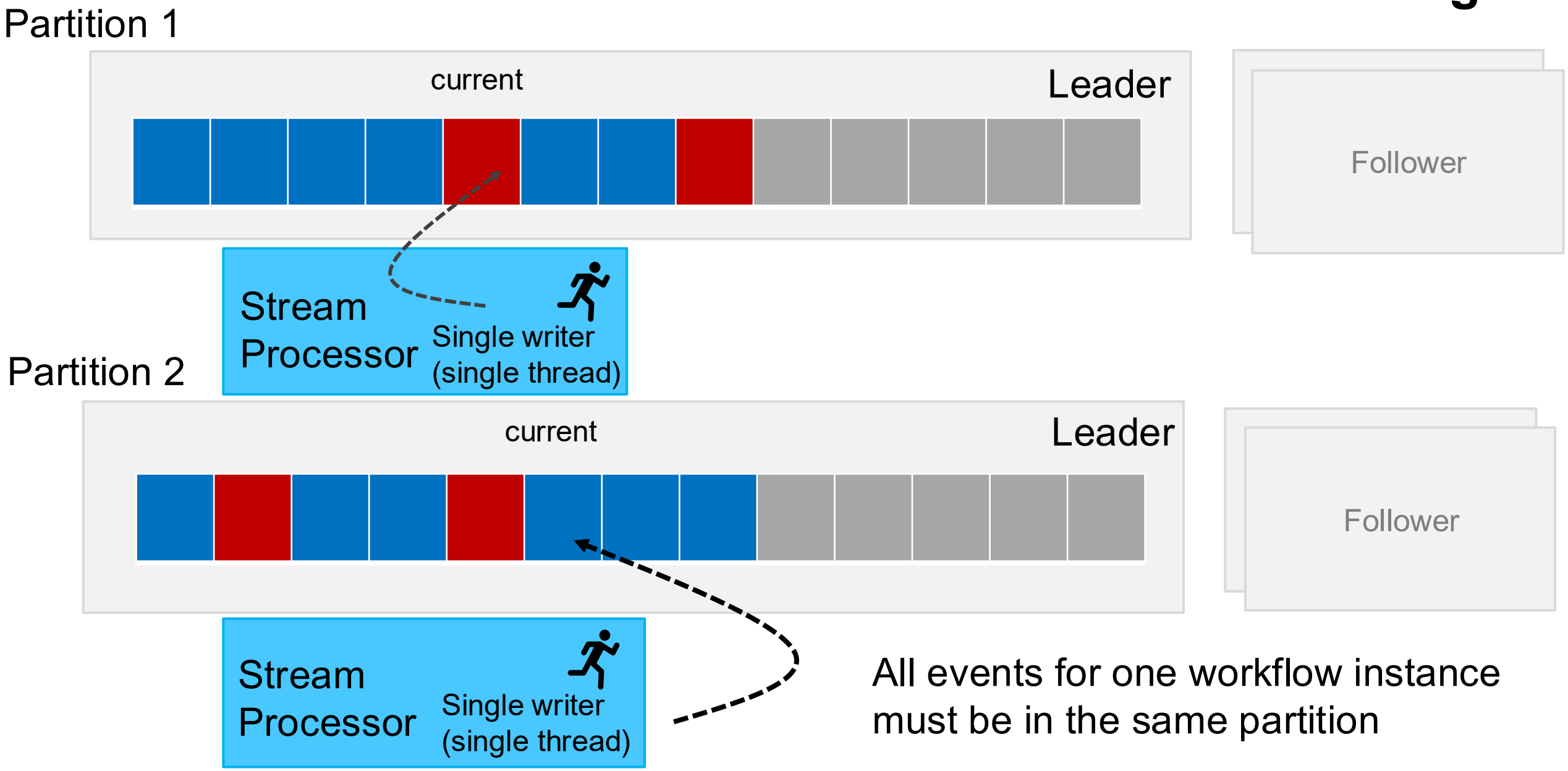
Zeebe Architecture



Peer-to-Peer Clusters and Replication

- To provide fault tolerance and resilience multiple Zeebe brokers form a cluster
- When log entries are written by the leader, they are first replicated to the followers before they are executed
  - Guarantees that every log entry that gets processed is correctly replicated
  - Replication guarantees that no committed entry gets lost
- Clients connect to the gateway which knows the topology of the Zeebe broker cluster

Partitioning



All events for one workflow instance must be in the same partition

Partitions

- All events related to one workflow instance must go onto the same partition to ensure the single writer principle
  - current state in a broker node can be restored locally
  - Round robin mechanism to decide onto which partition an instance goes
  - The partition even becomes part of the workflow instance id

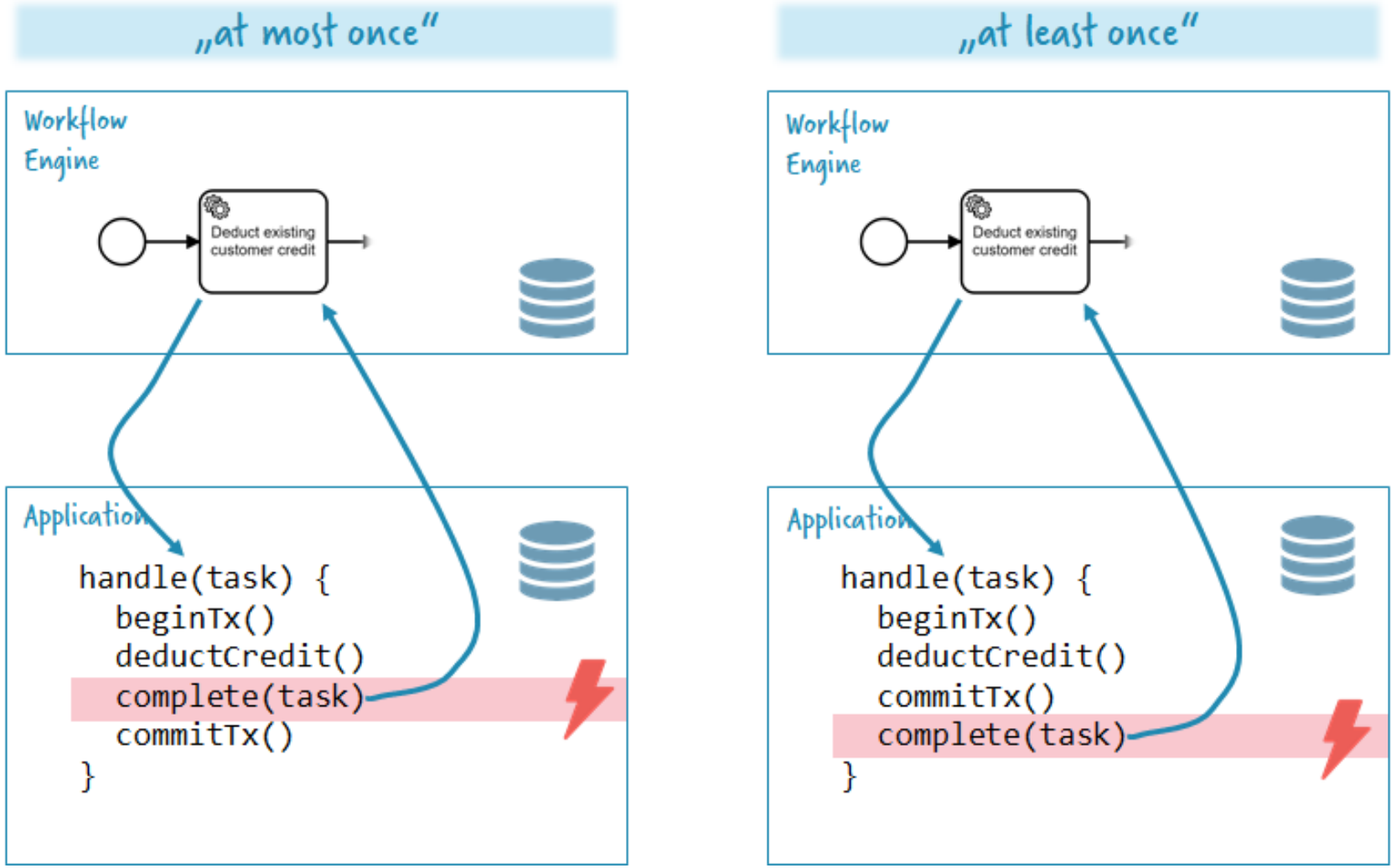
- Message Correlation: a workflow instance might wait for messages (or events) to arrive
  - Zeebe creates a message subscription which lives on a partition that might be different to the one where the workflow instance runs
  - The partition is determined using a hash function on the correlation identifier and can thus be easily found

Transactions and At-least Once Semantics

Note: Zeebe clients do not implement any form of ACID transaction protocols.

Design Option 1 (at most once): you inform Zeebe about job completion and then you commit the transaction to your domain

Design Option 2 (at least once): you commit the transaction to your domain and then you inform Zeebe about job completion



Source: https://blog.bernd-ruecker.com/zeebe-io-a-horizontally-scalable-distributed-workflow-engine-45788a90d549

Queries via CQRS

- The Zeebe Broker is responsible for executing running workflows and is optimized to apply commands to the current state in a way that performance and scalability goals are reached
- Due to the lack of a relational database, queries cannot be done using a simple SELECT statement
- Need for a separate query model (also denoted as CQRS)
- Zeebe provides exporters (e.g., to Elasticsearch) and provides an operation tool called Operate to look into running process instances

Camunda Operate

Dashboard | Running Instances 667 | Filters 62 | Incidents 581 | Selections 0 0 | Trial License

Filters 62

Flight registration

Version 2

Instance Id(s) separated by space or comma

Error Message

Start Date

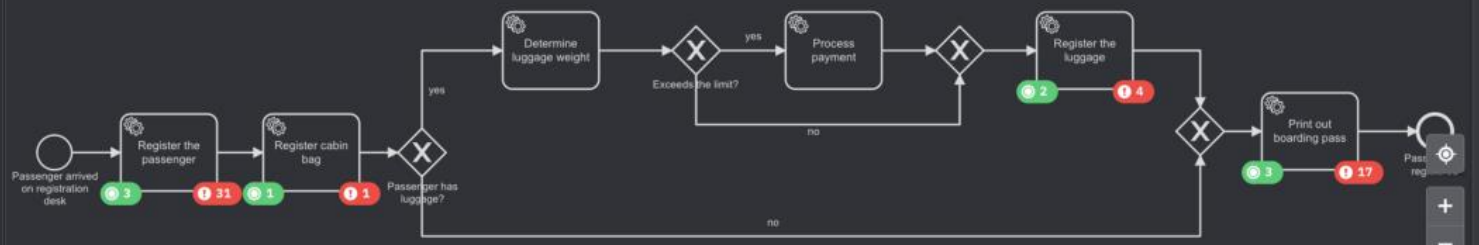
End Date

Flow Node

- [x] Running Instances
  - [x] Active
  - [x] Incidents
- [ ] Finished Instances
  - [ ] Completed
  - [ ] Canceled

Reset Filters

Flight registration



Selections 0

Instances

Workflow	Instance Id	Version	Start Time	End Time	Actions
Flight registration	4877	Version 2	3 Apr 2019 17:08:51	--	
Flight registration	4885	Version 2	3 Apr 2019 17:08:51	--	
Flight registration	4926	Version 2	3 Apr 2019 17:08:53	--	
Flight registration	5029	Version 2	3 Apr 2019 17:08:54	--	
Flight registration	5074	Version 2	3 Apr 2019 17:08:55	--	
Flight registration	5099	Version 2	3 Apr 2019 17:08:55	--	

Create Selection

1 2 3 4 5 ... 11

Screenshot of Operate

Some Nice Recent Articles

CloudEvents verstehen und nutzen: Der CNCF-Standard für Event-Integration

CloudEvents schafft Interoperabilität für Event-basierte Systeme. Warum der Standard sinnvoll ist und wie er funktioniert, behandelt dieser Blogpost.

(Bild: Shutterstock/Milos Milosevic)

10:28 Uhr Lesezeit: 8 Min. | Developer

Von Golo Roden

Event Sourcing: Die bessere Art zu entwickeln?

Event Sourcing ist ein alternativer Ansatz für das Speichern und Verwalten von Daten. Wie funktioniert Event Sourcing und was sind die Vor- und Nachteile?

23

(Bild: sabthai/Shutterstock.com)

29.01.2025, 10:43 Uhr Lesezeit: 29 Min. | Developer

Von Golo Roden

CQRS als Grundlage für moderne, flexible und skalierbare Anwendungsarchitektur

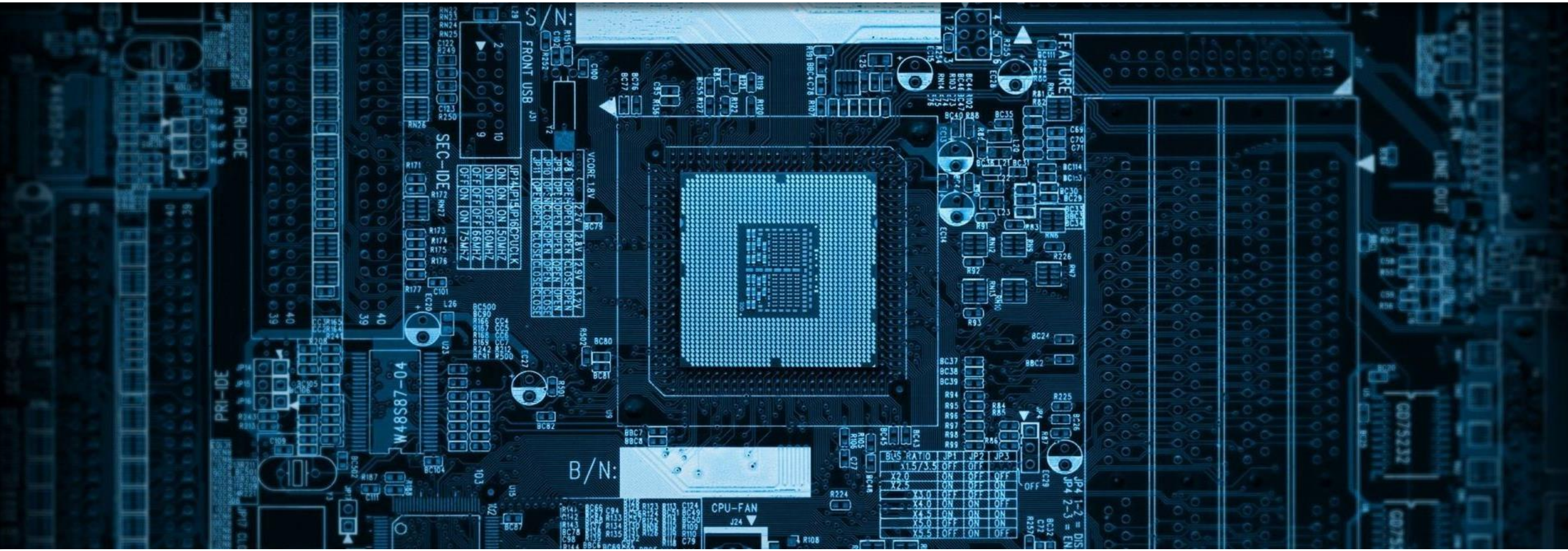
Man hört oft, komplexe Systeme basierten auf CQRS. Doch was genau ist das eigentlich, CQRS? Was verbirgt sich hinter dem Akronym, und wie kann man es nutzen?

13

(Bild: erzeugt mit KI durch iX)

12.02.2025, 09:02 Uhr Lesezeit: 26 Min. | Developer

Von Golo Roden



Questions?